

①

ML, NLP, PL Bootcamp

Brush Neck SM

Sox - 20

Lex - 12 Summary

Summary:-
Descriptive
Statistics

Lex - 13 Correlation
& Covariance

Summary Summary:-

- ① Minimum
- ② 1st Quartile (25%)
- ③ Median
- ④ 3rd Quartile (75%)
- ⑤ Maximum

Ex:-

1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7,
8, 8, 9, 2

[Lower Fence Value ← → Higher Fence Value]

Anything lower than
Lower fence is an
outlier

Also anything higher
than Higher Fence is
also an outlier

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

↙
Quartile 1

$$\text{Higher fence} = Q_3 + 1.5(IQR)$$

↙
Quartile 3

Q_1 (Quartile 1)

$$25^{\text{th}} \text{ Percentile} = \frac{\text{Percentile}}{100} \times (n+1) \quad n=19$$

$$= \frac{25}{100} \times (19+1)$$

$$= \frac{25}{100} \times 20$$

$$5^{\text{th}} \text{ position} = 3$$

$$Q_1 = 5^{\text{th}} \text{ position} = 3$$

(3)

Now calculate Q_3

(Q_3) 75%ile is

$$Q_3 = \frac{75}{100} \times (79+1)$$

$$= \frac{3}{4} \times 26^5 = 15^{\text{th}} \text{ position}$$

15th position = 7

$$IQR = Q_3 - Q_1 = 7 - 3 = 4 \quad \text{✓}$$

$$\begin{aligned} \text{Lower fence} &= Q_1 - 1.5(IQR) \\ &= 3 - 1.5(4) \\ &= 3 - 1.5(4) \\ &= 3 - 6 \\ &= -3 \end{aligned}$$

$$\text{Lower fence} = 3$$

$$\begin{aligned} \text{Upper fence} &= Q_3 + 1.5(IQR) \\ &= 7 + 1.5(4) \\ &= 7 + 6 \\ &= 13 \quad \text{Ans} \end{aligned}$$

Here we take the
element 27 as its an
outlier.

⇒ Now write the distribution again by removing 2)

Ex 1 - 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6,
7, 8, 8, 9, ~~10~~

Minimum = 1

Q_1 (1st Quartile) = 3

$$\text{Median} = \frac{5+5}{2} = \frac{10}{2} = 5$$

Q_3 (3rd Quartile) = 7

Maximum = 9

→ This is the whole Five number summary.

Using this we can plot the box plot.

⇒ Covariance & Correlation

Covariance and Correlation are two statistical measures of relationship between two variables.

Both are used to understand changes in / or relationship between variables.

⇒ Covariance :- It is a measure of how much two random variables change together. If the variables tend to increase or decrease together then covariance is positive.

If one tends to increase while the other decreases then covariance is negative.

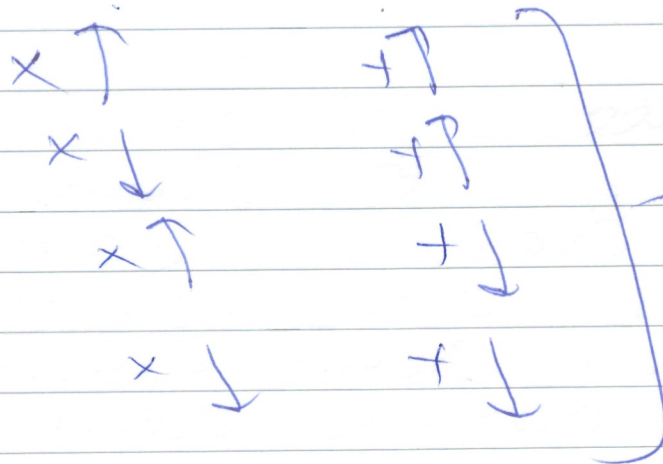
=>

Suppose we have
values in 2 columns

X	Y
2	3
4	5
6	7
8	9

Now if we quantify
the relationship between

X & Y



4 possible

Suppose we have a
dataset as

Size of
House

Price

1200

45000

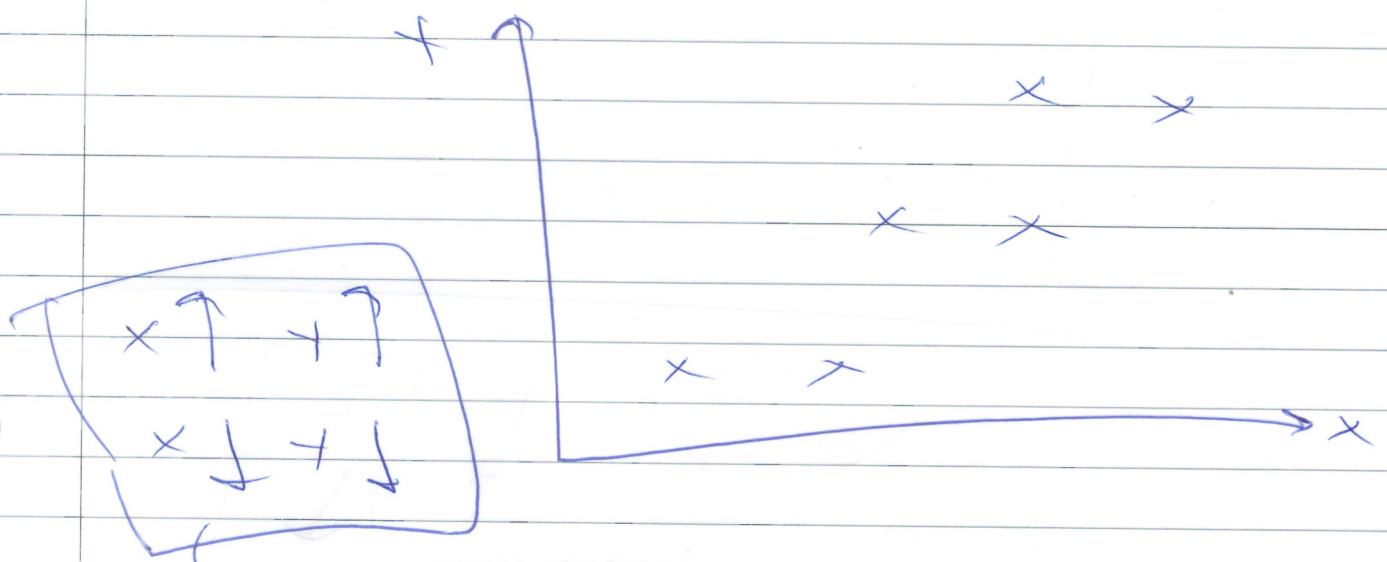
1300

50000

1500

75000

Now we plot the
points as

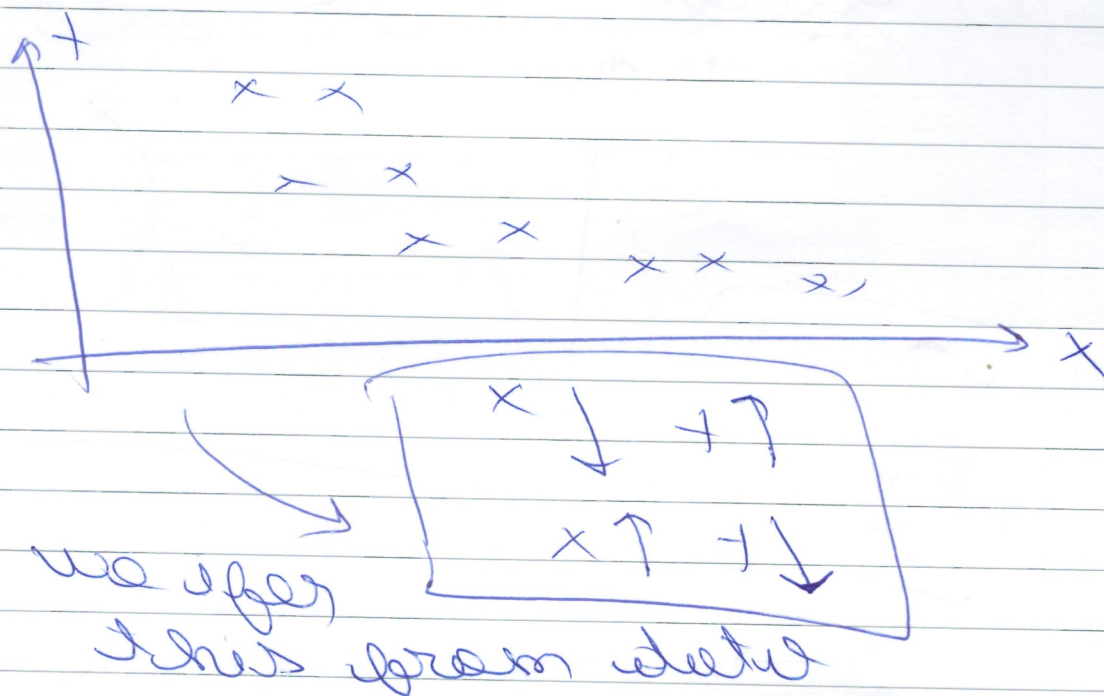


These are our test set

Now we take another
job as example:

X	Y
7	10
6	12
5	14
4	16

If we draw the graph



⇒ Now if we go with covariance formula

$$\Rightarrow \text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

→ Now if we calculate the covariance of x with x

$$\text{Cov}(x, x) = \sum_{i=1}^n \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$$

$$\text{Cov}(x, x) = \text{Var}(x)$$

→ This is basically the variance only

$$\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

x_i → Data point of random variable x

$\bar{x}$ → Sample mean of x

$y_i \rightarrow$ data points of
random variable

$\bar{y} \rightarrow$ sample mean of y

$\Rightarrow$ we take the sample
of data y'

Students:-

Hours studied

Exam
Score

2

50

3

60

4

70

5

80

6

90

$\times \uparrow \uparrow$

$\times \downarrow \downarrow$



↑ve covariance

(11)

11000 x is directly proportional to y

$$\begin{aligned} \textcircled{1} \quad \bar{x} &= \frac{2+3+4+5+6}{5} \\ (\text{mean}) &= \frac{20}{5} = 4 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad \bar{y} &= \frac{50+60+70+80+90}{5} \\ (\text{mean}) &= 70 \end{aligned}$$

$$\text{Cov}(x, y) =$$

$$(5-4)(80-70) + (6-4)(90-70)$$

$$\frac{(2-4)(50-70) + (3-4)(60-70) + (4-4)(70-70) + (5-4)(80-70) + (6-4)(90-70)}{(5-1)}$$

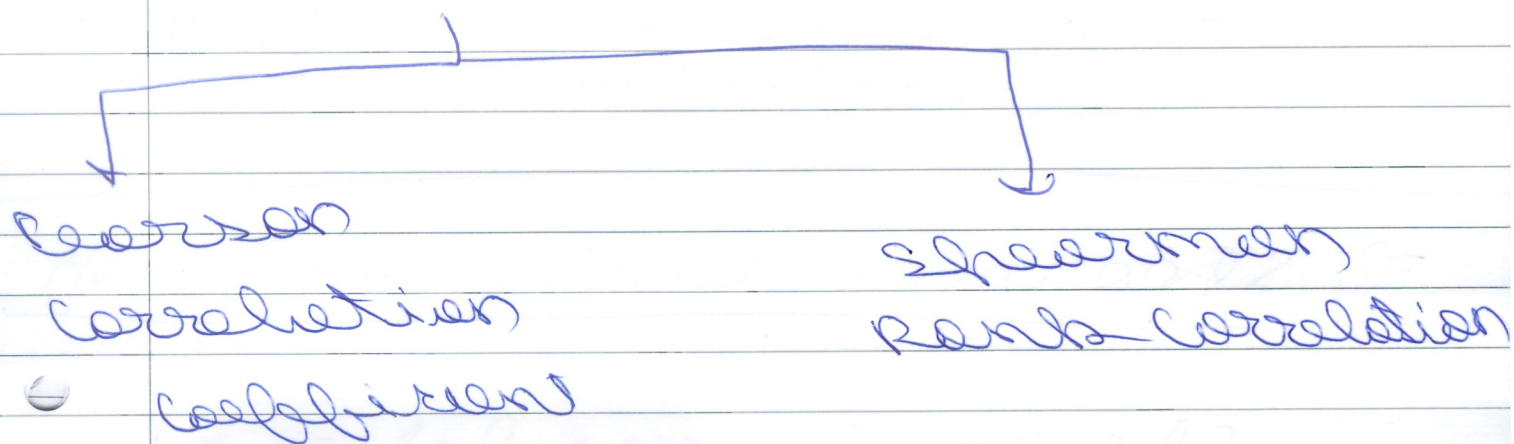
$$\text{Cov}(x, y) = 20$$

The positive covariance indicates that the no. of hours studied increases the exam score also.

⇒ Advantages & Disadvantages of Covariance

<u>Advantages</u>	<u>Disadvantages</u>
① Quantify the relationship between two variables	① Covariance does not have any specific limit value ↓ As the var (x) → ∞ or 0 it can go from +∞ to -∞

⇒ Correlation



(1) Pearson Correlation Coefficient -

↳ Range from [-1 to 1]

$$\rho_{x,y} = \frac{\text{cov.}(x,y)}{\sigma_x \cdot \sigma_y}$$

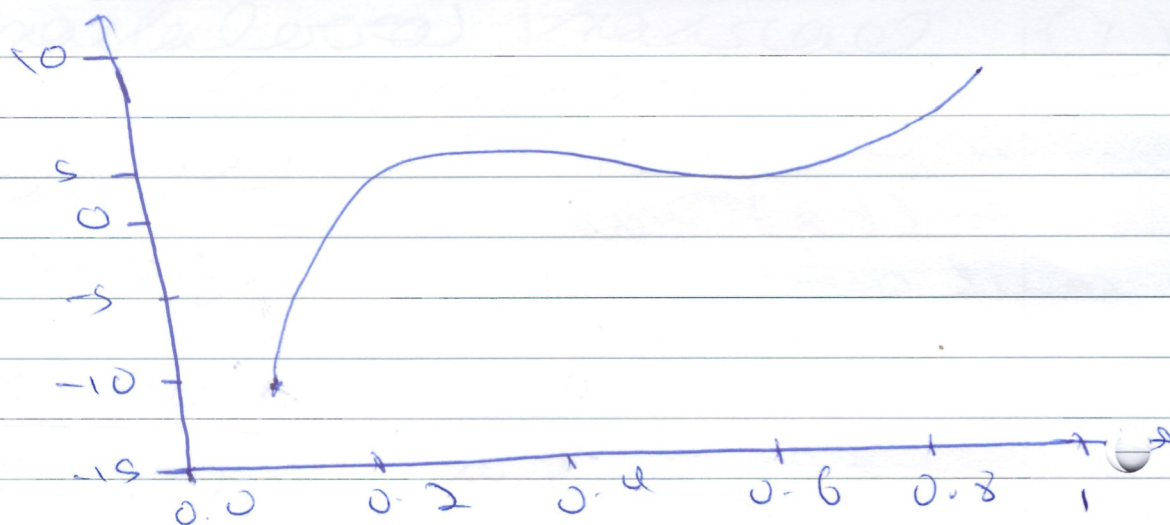
↙ Pearson Correlation
 ↘ Standard Deviation of x & y

(2) The more the correlation towards +1, the more correlated x & y is

The more the value
towards -1, the more
-ve ~~correlated~~ is X & Y;
correlated

⇒ Spearman Rank correlation:

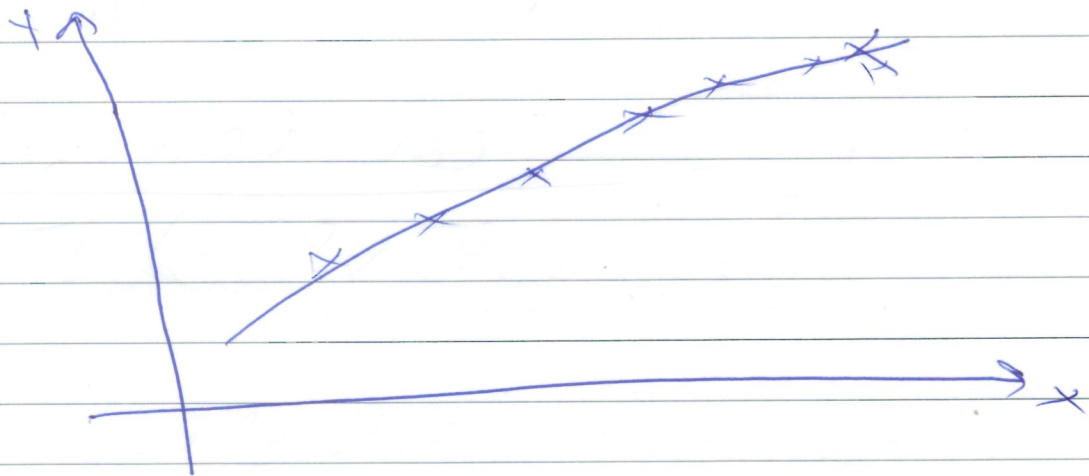
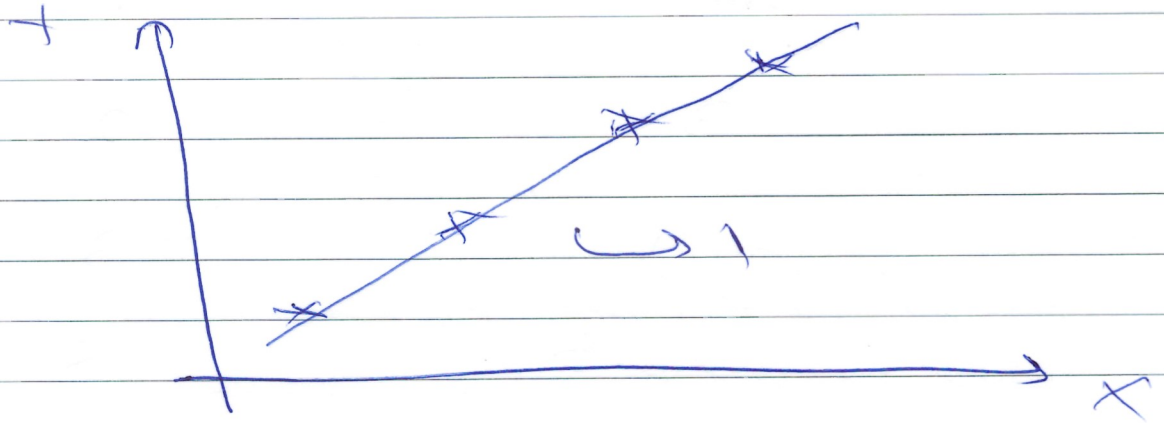
Spearman correlation = 1
Pearson correlation = 0.88



$\begin{matrix} \uparrow & \uparrow \\ X & Y \\ \downarrow & \downarrow \end{matrix}$

→ Pearson correlation = 0.8

Pearson correlation coefficient for non linear data



$$r_s = \frac{\text{Cov}(R(x), R(y))}{\sigma(R(x))\sigma(R(y))}$$

x	1
1	2
3	4
5	6
7	8
9	7

$R(x)$

$R(y)$

2

1

3

2

4

3

5

5

1

4

$$r_s = \frac{\text{Cov}(R(x), R(y))}{\sigma(R(x)) \cdot \sigma(R(y))}$$

Feature selection

Size of House

No. of Room

Location

Price

~~Best~~ Best

Household

Price